LINUX SYSTEMS ADMINISTRATION

PRACTICAL LAB PORTFOLIO

Ubuntu Server Troubleshooting

Permissions • systemd • Storage • SSH

A documented, evidence-led lab report showing investigation, remediation, and verification on an Ubuntu Server virtual machine.

Environment
Ubuntu Server 26.04.1 LTS
Hostname: server1
User: samuel
VirtualBox NAT network

PORTFOLIO OVERVIEW

This portfolio records four practical Linux support scenarios using command-line investigation and direct verification. The supplied terminal screenshots are embedded as evidence; Windows-only material in the screenshot pack is intentionally excluded.

SCENARIO	RESULT	STATUS
File permissions & access control	Restricted a payroll file to its owner and authorised group; verified permitted access and append operation.	**Completed**
systemd service failure	Diagnosed an ExecStart executable-path failure (203/EXEC), corrected the command, and verified successful one-shot execution.	**Completed**
Disk usage & log investigation	Measured a controlled 200 MB file impact, identified large log consumers, removed test data, and verified disk usage returned to baseline.	**Completed**
SSH diagnostics	Checked listener, network address, host-key prompt and successful localhost authentication; no SSH repair was needed.	**Diagnostic**

Scope note. The APT package/dependency exercise was not completed and is not presented as a finished case study. This report distinguishes demonstrated outcomes from work that remains outside scope.

Environment & Method

Lab environment

Component	Observed configuration
Operating system	Ubuntu Server 26.04.1 LTS
Virtualisation	Oracle VirtualBox
Hostname / account	server1 / samuel
Network interface	enp0s3, address 10.0.2.15/24; VirtualBox NAT
SSH transport	TCP port 22, IPv4 and IPv6 listeners observed

Troubleshooting workflow

- **Establish state:** collect command output and service, filesystem, or access status before changing configuration.
- **Isolate the cause:** correlate symptoms with the relevant logs, ownership/mode bits, or service execution details.
- **Apply a bounded change:** alter only the setting or test artefact needed for the scenario.
- **Verify:** re-run the relevant command or test and record the resulting state.

Evidence standard

Screenshots are included as submitted. A screenshot demonstrates only what is visible in it; explanations in this report also reflect the lab outcomes recorded during the work. No claim is made that this was a production incident or a real outage.

File Permissions & Access Control

Objective

Apply least-privilege access to a payroll report and verify that authorised group membership enables the required file operations while unauthorised access remains restricted.

Target and configuration

```
/srv/company/Payroll/report.txt
Owner: root Group: Payroll Mode: 0660 (-rw-rw )
Directory: /srv/company/Payroll (group Payroll; setgid directory)
```

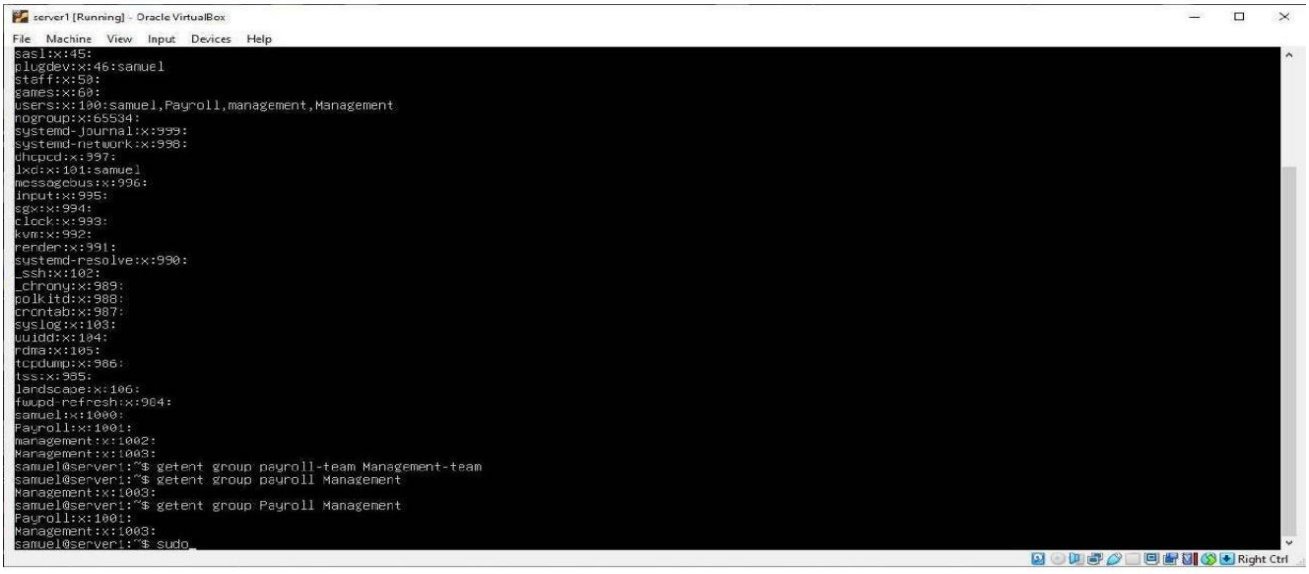
Investigation and action

- Checked the relevant group and user identities with `getent` and `id`.
- Inspected directory and file ownership/mode using `ls -ld` and `ls -l`.
- Used the Payroll account to read the report and append a test entry; verified the appended line.
- The lab outcome recorded for the Management account was access denied, as intended by the permissions model.

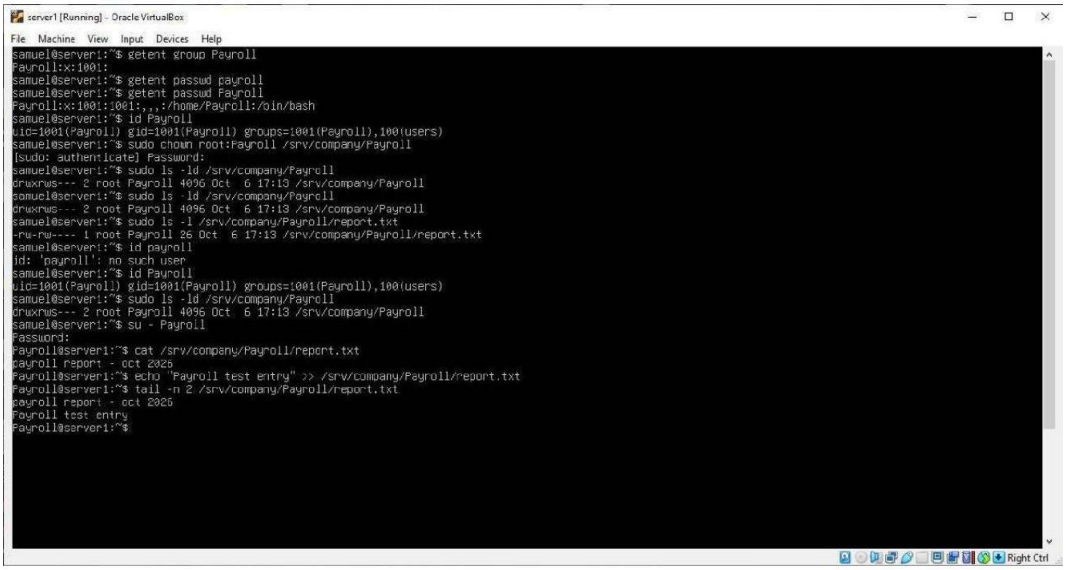
Outcome

The file is readable and writable by the owner and members of the Payroll group, while access for other users is not granted by the file mode. The directory setgid bit supports consistent group inheritance for new items created there.

Evidence



Evidence 2.1 — Group and account inspection before access testing.



Evidence 2.2 — Payroll account reads and appends a test entry to the report.

systemd Service Failure

Symptom

The custom lab unit `lab-demo.service` failed to start. systemd reported an execution failure with `status=203/EXEC`. The journal identified the configured executable path as unavailable.

Root cause

The unit's `ExecStart` referenced a command path that did not exist: `/usr/local/bin/lab-command-that-does-not-exist`. The service manager could not execute the configured program.

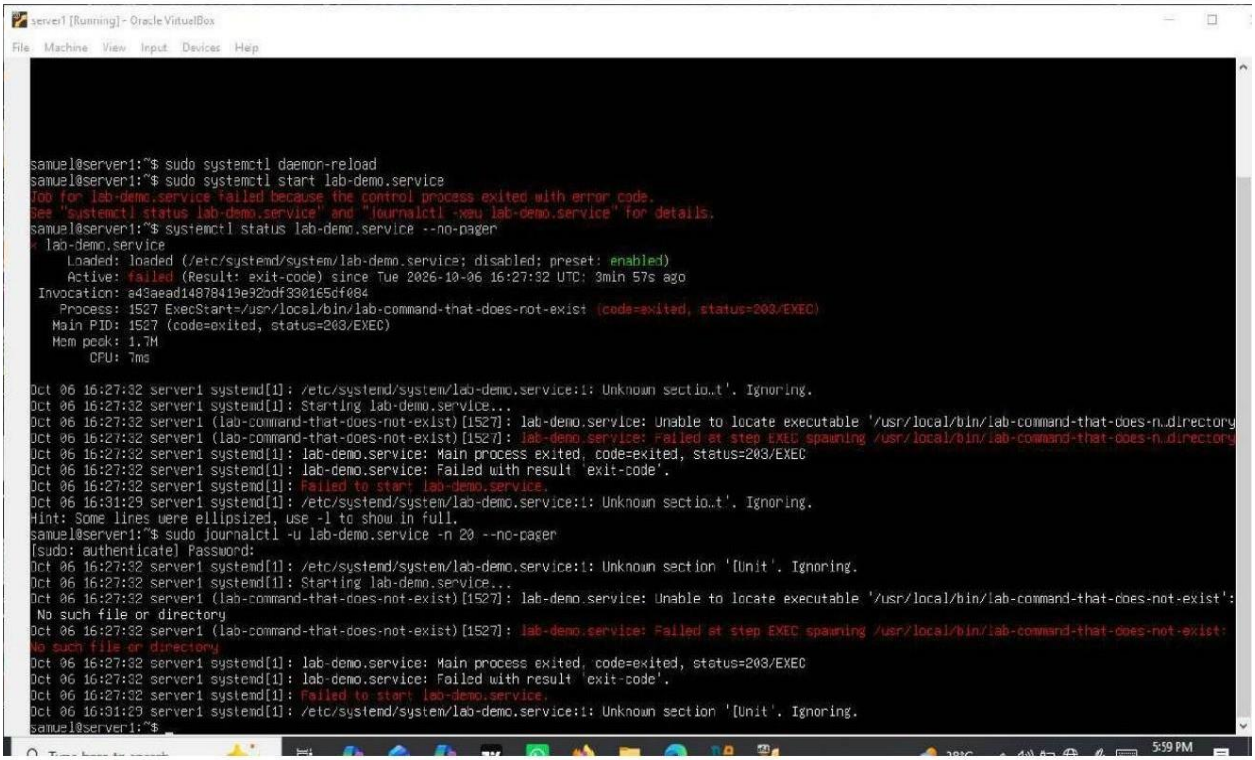
Remediation

```
sudo systemctl daemon-reload
sudo systemctl start lab-
demo.service systemctl status
lab-demo.service --no-pager
sudo journalctl -u lab-demo.service -n 20 --no-pager
```

The unit was updated to execute `/usr/bin/touch /tmp/lab-demo-success`, then reloaded and started again. The marker file was used to verify the one-shot action.

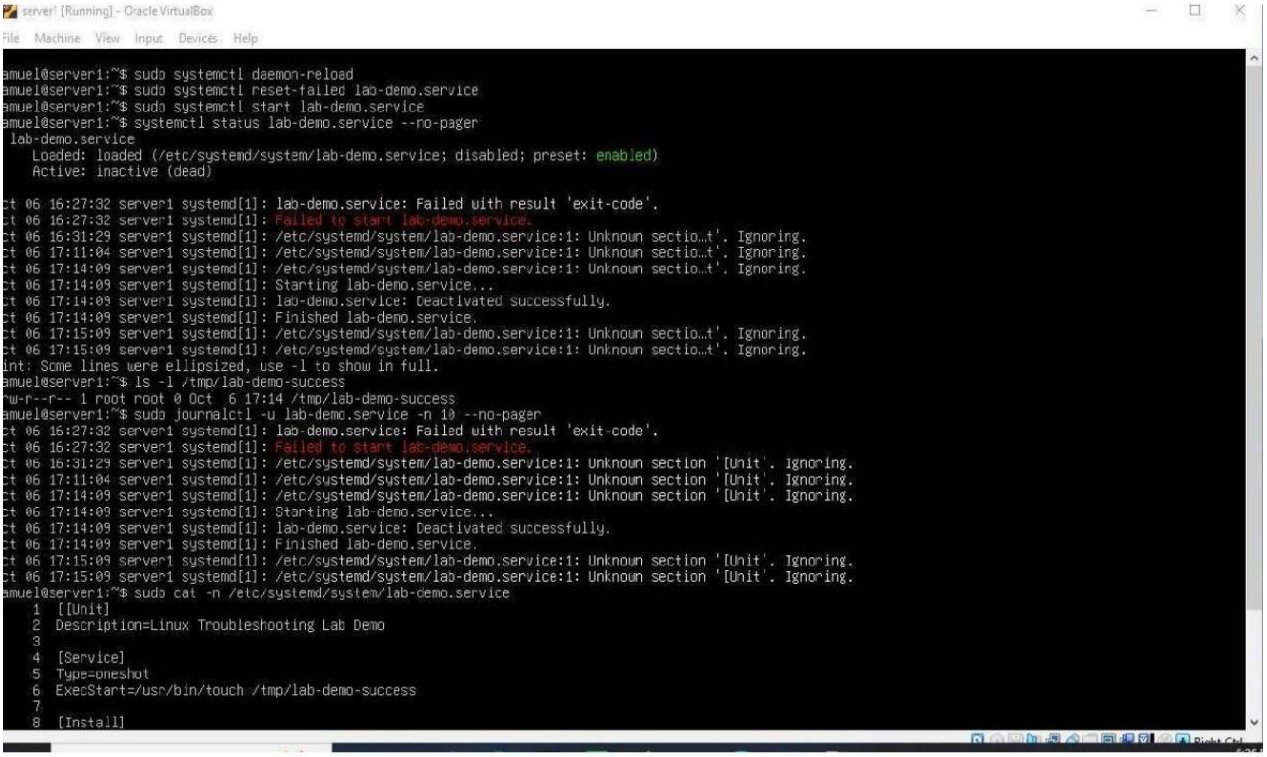
Verification and interpretation

- Initial evidence shows a failed unit and an executable lookup failure.
- The corrected command completed successfully and created the marker file.
- For a `Type=oneshot` unit without a persistent process, `inactive (dead)` after a successful run can be normal; the result and marker file are more useful than expecting a continuously active state.



Evidence | systemd Service Failure

Evidence 3.1 — Failed unit status and journal output identify the ExecStart path failure.



Evidence 3.2 — Additional unit diagnostics captured during investigation; not treated as the primary cause without supporting evidence.

Disk Usage & Log Investigation

Objective

Measure filesystem utilisation, identify log-directory consumers, model a controlled increase in disk usage, remove the test artefact, and verify recovery. This was a lab simulation—not a real disk-full outage.

Baseline and inspection

```
df -h
sudo du -sh /var/log/lab-demo
sudo du -xhd1 /var/log 2>/dev/null | sort -h
```

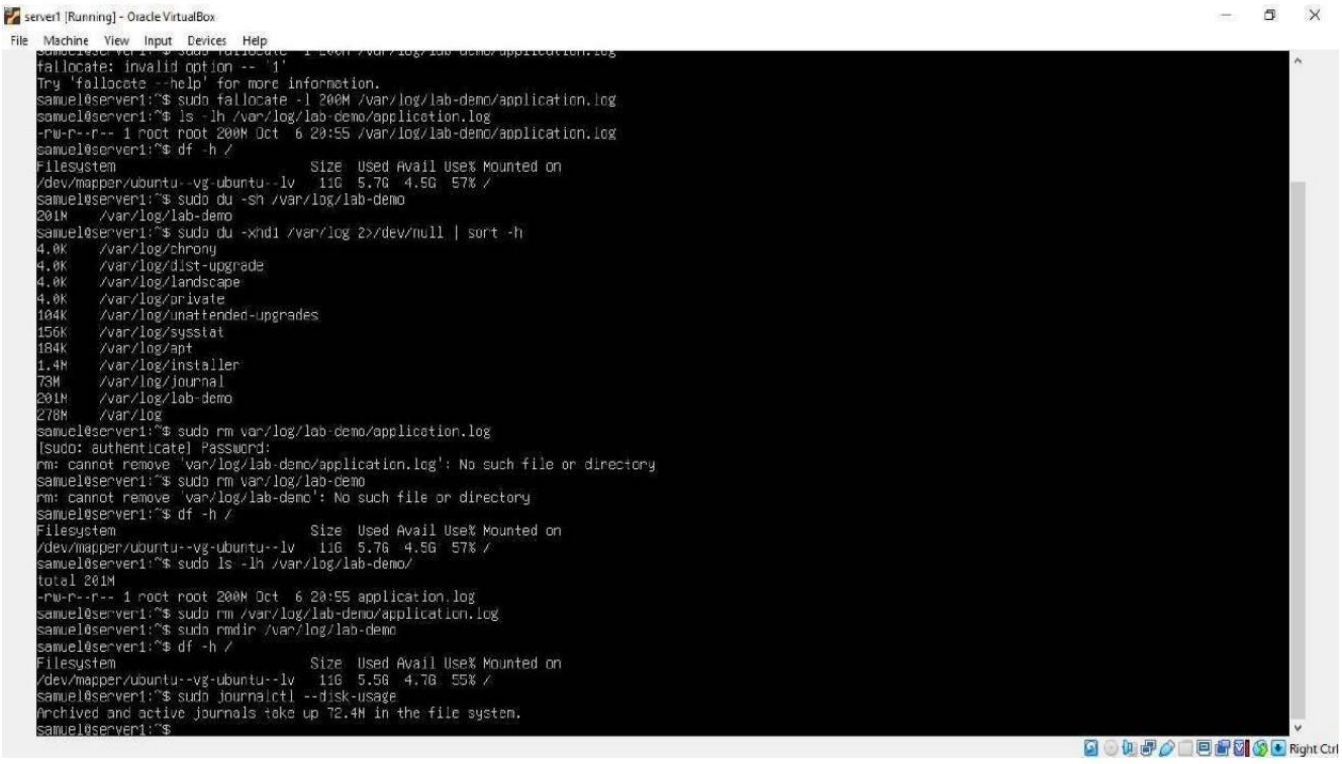
The baseline root filesystem was approximately 11 GB, with about 4.7 GB available and utilisation around 55%. The journal accounted for roughly 72.4 MB; directory-size output was used to identify where log storage was concentrated.

Controlled test and cleanup

```
sudo fallocate -l 200M /var/log/lab-
demo/application.log ls -lh /var/log/lab-
demo/application.log
df -h
sudo rm /var/log/lab-
demo/application.log sudo rmdir
/var/log/lab-demo
df -h
```

Observed result

- The 200 MB test file increased root filesystem utilisation from approximately 55% to 57%.
- An initial cleanup attempt used an incorrect path/sequence and returned “No such file or directory”; the file and directory were then removed with the correct commands.
- The final `df -h` output showed approximately 4.7 GB available and 55% usage again, consistent with the baseline.



Evidence 4.1 — Controlled 200 MB file, filesystem usage increase, cleanup, and return to 55% usage.

SSH Diagnostics & Access Verification

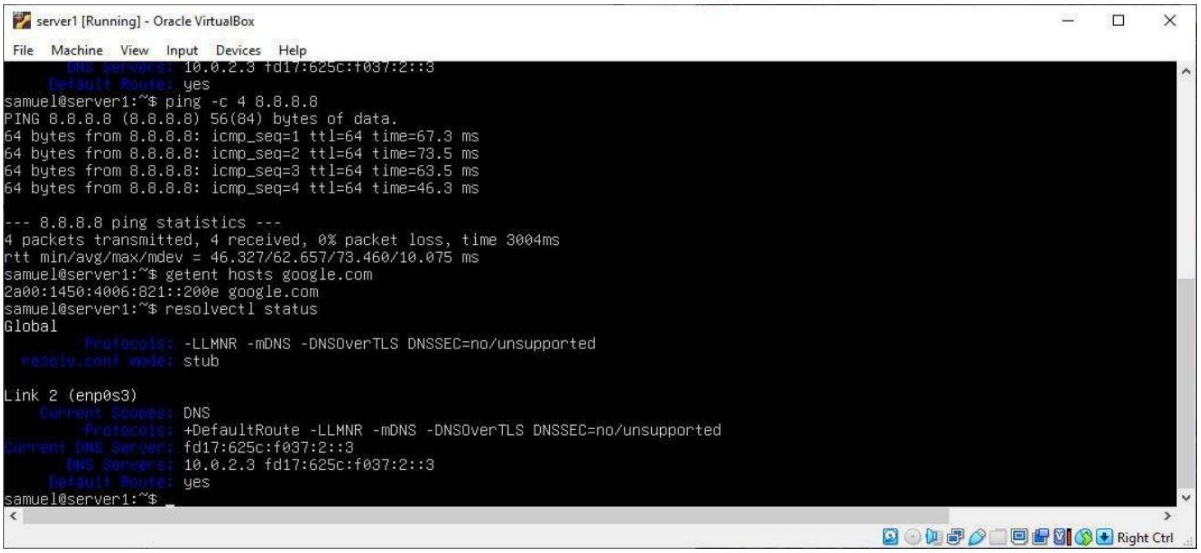
Objective

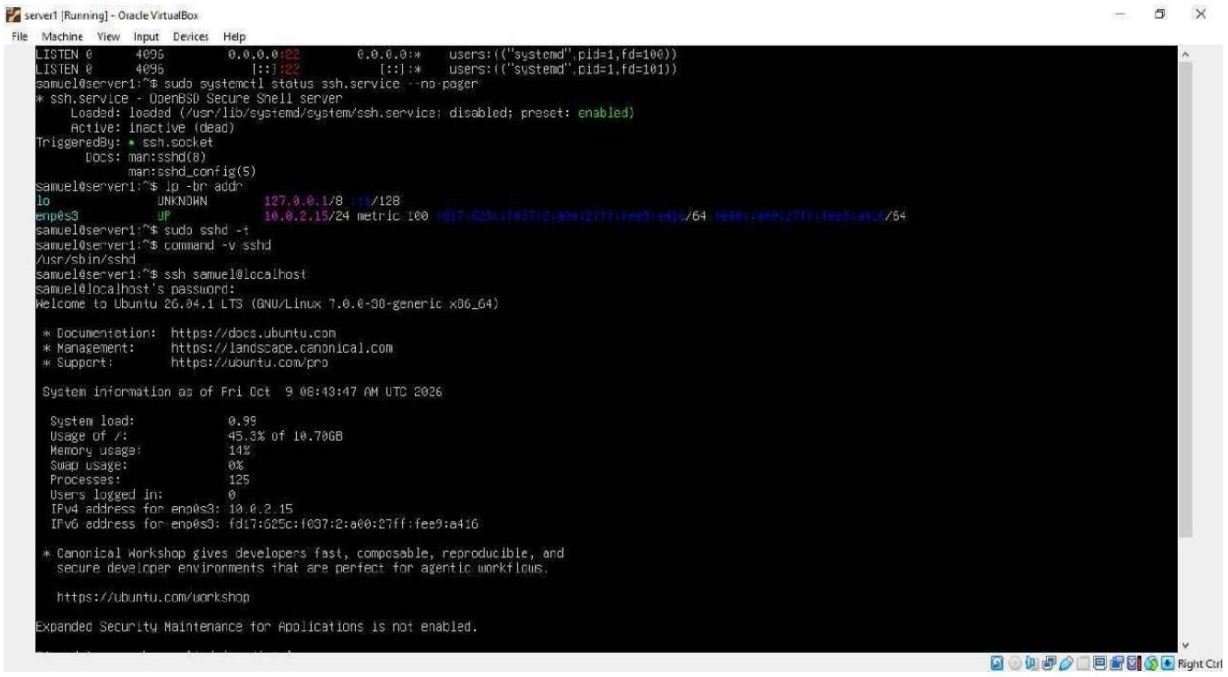
Confirm that SSH is listening, identify the server address, understand socket-activated service state, and test local authentication.

Findings

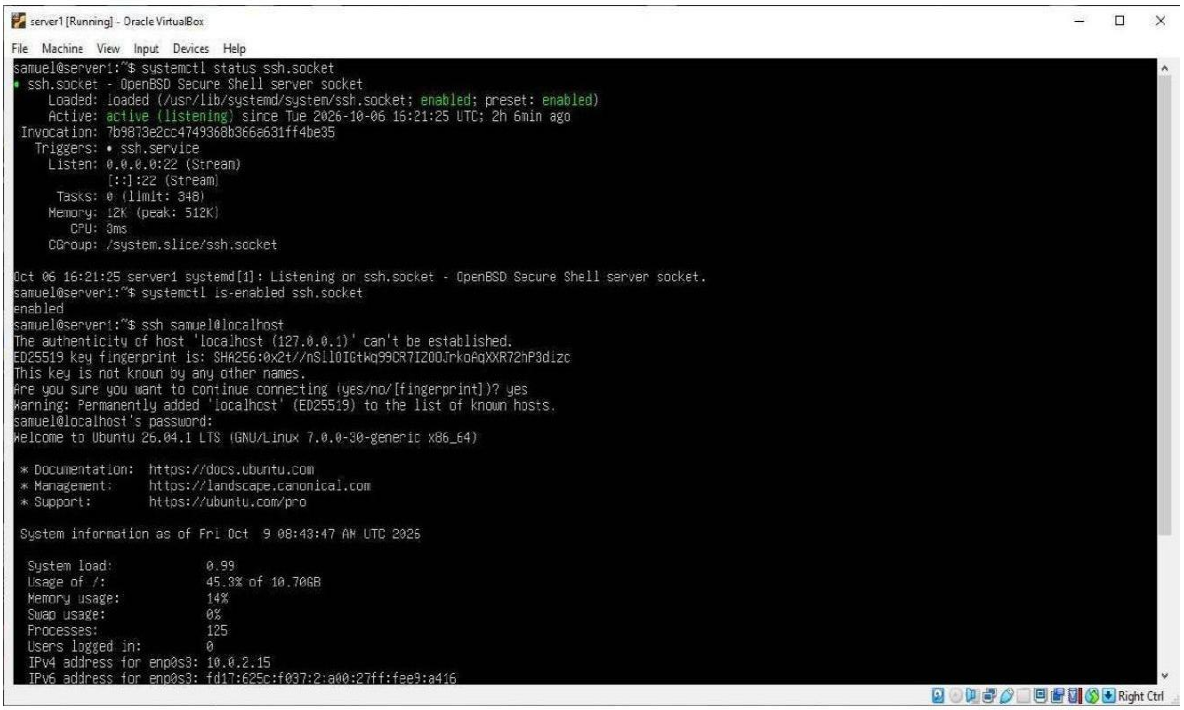
- The network interface `enp0s3` was up with IPv4 address `10.0.2.15/24` on the VirtualBox NAT network.
- TCP port 22 was listening on IPv4 and IPv6.
- `ssh.socket` was enabled/listening. The associated `ssh.service` could appear inactive before a connection because socket activation starts the service when needed.
- A local connection using `ssh samuel@localhost` succeeded; the journal recorded successful password authentication and session creation.

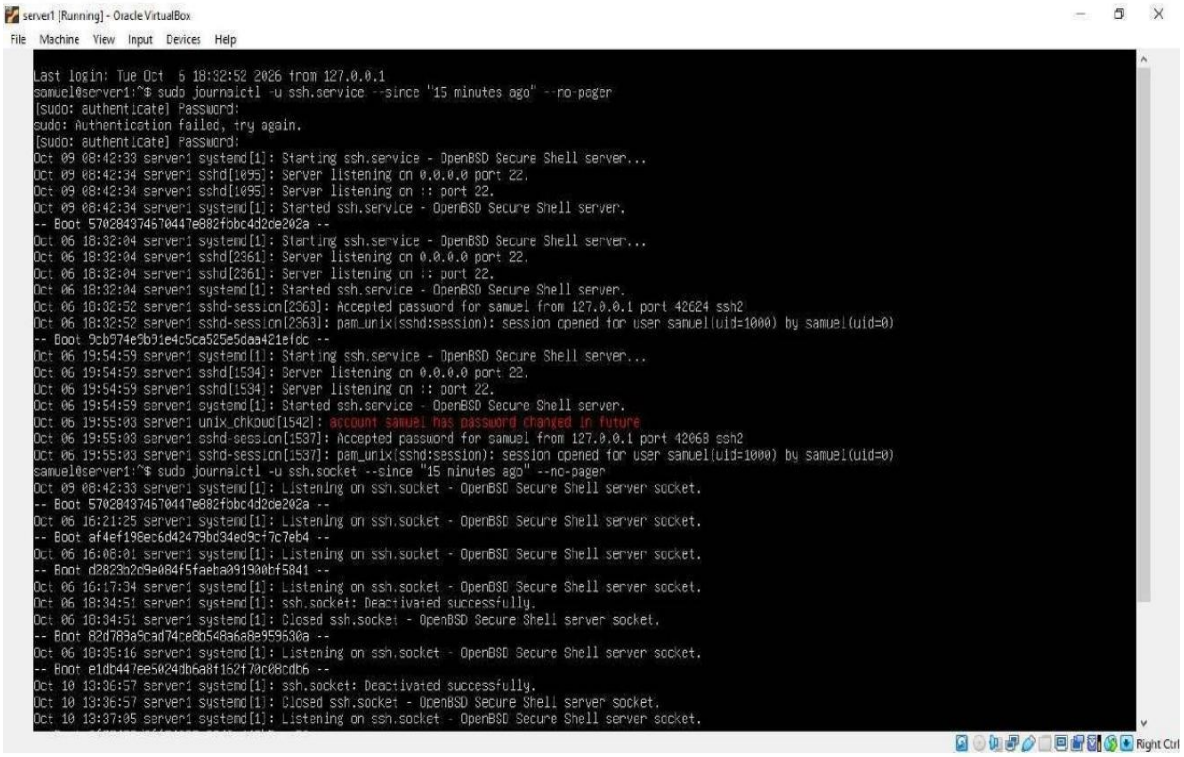
SSH Diagnostics & Access Verification





Evidence | SSH Connection Verification





Reusable Command Reference

System and network triage

```
hostnamectl
ip -br addr
ip route
ss      -lntp
sudo sshd -t
```

systemd triage

```
systemctl status <unit> --no-pager
sudo journalctl -u <unit> -n 50 --no-pager
systemctl cat <unit>
sudo systemctl daemon-reload
```

Storage triage

```
df -h
df -i
sudo du -xhd1 /var/log 2>/dev/null | sort -h
sudo journalctl --disk-usage
```

Permissions triage

```
id <user>
getent group <group>
ls -ld <directory>
ls -l <file>
namei -l <path>
```

Professional practice demonstrated

- Use evidence and logs to narrow the cause before changing configuration.
- Prefer the smallest corrective change and verify it explicitly.
- Distinguish a controlled simulation from a production incident.
- Capture baseline and post-change states, including cleanup confirmation.
- Report limitations honestly and avoid describing a healthy diagnostic check as a repair.

Scope and next improvements

The submitted screenshot pack also contained a Windows command-prompt image unrelated to these Linux cases; it was excluded. An APT dependency-conflict exercise was started separately but not completed, so it is not counted as a completed scenario. A future iteration could add a fully documented package dependency failure/recovery and a concise test matrix for user/group access